Polynomial Nuisance Profiling for Range–Inertial Initialization

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***Abstract*—We present a range–inertial initializer that turns an IMU-derived trajectory shape into inexpensive yaw candidates. Squared-range lifting and elimination of polynomial translation reduce candidate generation to a unit-circle problem, followed by robust raw-range refinement. We establish a sufficient recovery condition, prove invariance to polynomial displacement contamination, and characterize the tradeoff between bias absorption and heading information. At 30 m stand-off with a 3 m anchor height span, the profiled candidate reaches the best observed objective in all 200 paired trials, compared with 159 for a local yaw seed. It matches two uniform starts; on a sequential desktop timing subset for this geometry, median solve time decreases from 66.6 to 47.1 ms. Weak-height geometry identifies the tradeoff between additional starts and computation. Across 528 correlated configurations from 66 real windows, profiling reduces objective misses from 87 to 40, with smaller, recording-dependent improvements in endpoint position. These results provide an efficient initialization primitive and guide the choice of nuisance freedom in range–inertial localization.**

***Index Terms*—Ultra-wideband localization, inertial navigation, invariant extended Kalman filter, initialization, observability, robust estimation.**

I. Introduction

Two-way ranging (TWR) measures distance between an ultra-wideband (UWB) tag and a surveyed anchor. Initializing UWB–IMU localization requires resolving position and heading from these spherical observations while propagating body-frame inertial measurements. A useful initializer should reach a favorable solution basin with few nonlinear solves and expose the motion information supporting its heading estimate.

Invariant filtering structures navigation errors on a Lie group [1]. Oursland and Mehrabian applied it to IMU–UWB fusion with large heading errors, using time differences of arrival and a tag lever arm [2]. We retain invariant inertial tracking and develop an acquisition stage for a single tag, surveyed anchors, asynchronous TWR, and zero lever arm.

Two difficulties motivate this work. A local raw-range fit can select a poor yaw basin despite using a reasonable position seed. A short window can also fit ranges while inferring heading through an assumed accelerometer calibration: a direction constrained under fixed bias can become unobservable when bias is free.

Our approach combines a displacement known up to vertical rotation with an unknown translation polynomial. Eliminating linear nuisance variables from expanded ranges leaves two circular coordinates. The resulting yaw candidate and a common constant-velocity (CV) translation seed initialize the raw-range fit (Fig. 1).

The paper makes two contributions. First, we exploit the yaw-circle structure of an IMU-derived displacement to generate inexpensive candidates for a common robust range optimizer, and compare optimization reliability, physical errors, and computation with local and uniform multi-start policies. Second, we characterize how polynomial nuisance freedom absorbs acceleration mismatch while changing heading identifiability, through an invariance result, a yaw–bias ambiguity, and a rotation-dependent approximation bound.

II. Related Work and Scope

Pacholska, Dümbgen, and Scholefield recover trajectory-basis coefficients from asynchronous distances to known anchors using a polynomial or bandlimited trajectory and linear relaxation, with sufficient exact-recovery conditions [3]. Here the nonpolynomial inertial displacement is supplied, while its yaw and a translation polynomial are unknown. Exploiting this structure reduces the remaining candidate problem to a circle.

Goudar *et al.* estimate rigid-body pose and velocity from multiple tags using semidefinite relaxations under static or constant body-centric velocity models [4]. Those tag and motion constraints differ from our single-tag, accelerated inertial window. Trawny and Roumeliotis solve planar relative pose using ranges and known egomotion, with a global algebraic solution for their planar problem [5]. Molina Martel *et al.* recover a four-degree-of-freedom relative pose from ranges and visual-SLAM trajectories under their distance and motion conditions [6]. Our input is an IMU-derived shape with unknown world velocity and possible polynomial contamination. This structure supports the circle reduction and the nuisance-invariance analysis developed below.

Inertial symmetries and bias-dependent observability are established topics [1], [7], [8]; our analysis specializes them to the window's nuisance directions. Unknown-landmark equivariant SLAM [9] has additional gauge freedoms absent with surveyed anchors.

Pfeiffer *et al.* provide efficient UWB moving-horizon estimation and the dataset used here [10], [11], with subsequent estimator improvements [12]. Jiang *et al.* combine TWR and IMU data using a ten-second factor-graph initialization and incremental smoothing [13]. These systems address complete navigation, while our controlled comparisons isolate candidate generation within a shared window optimizer and invariant tracker.

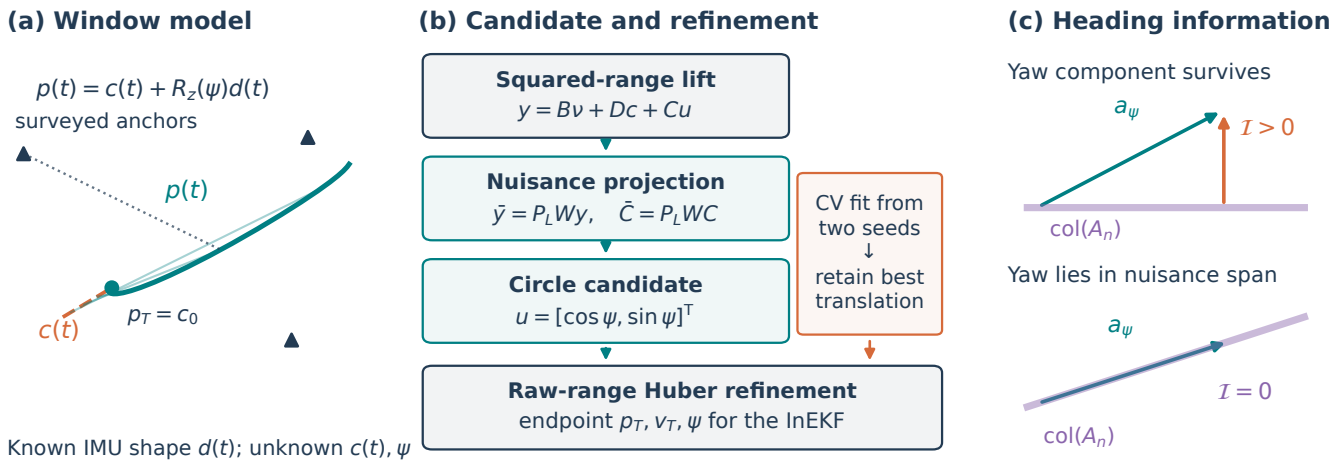


Fig. 1. Method overview. (a) An IMU shape rotated by unknown yaw augments polynomial translation. (b) Nuisance projection reduces candidate generation to a circle search; the selected yaw and a common CV translation seed initialize local raw-range refinement. (c) Heading information is the yaw-sensitivity component outside the nuisance span. The geometry and projection sketches are schematic.

III. Problem Formulation

Let $a_i \in \mathbb{R}^3$ be a surveyed anchor and let the bias-corrected range at its actual transaction time be

$$r_i = z_i - \beta_i = \rho_i + e_i, \qquad \rho_i = \|p(t_i) - a_i\|. \tag{1}$$

The tag lever arm is zero. Independent Gaussian $e_i \sim \mathcal{N}(0, \sigma_i^2)$ is used only for the inlier noise calculation; contaminated and real measurements need not follow that model. The anchor bias β_i and scale σ_i are fixed before an evaluation replay.

The nominal inertial model is

$$\begin{aligned} \dot{R} &= R[\omega_m - b_g]_\times, & \dot{p} &= v, \\ \dot{v} &= R(f_m - b_a) + g, & g &= -g_0 e_3, \end{aligned} \tag{2}$$

with $g_0 = 9.81\,\mathrm{m/s^2}$. Static acceleration constrains the gravity direction but does not determine yaw. Over a short window, first assume known biases and write $R(t) = R_z(\psi)Q(t)$, where Q is propagated from gravity alignment with the measured gyroscope. With the final window time T as reference and $\tau = t - T$, reanchor the integrated displacement so that $d(T) = \dot{d}(T) = 0$. Since a vertical rotation leaves gravity unchanged,

$$p(t) = p_T + v_T\tau + R_z(\psi)d(t). \tag{3}$$

This identity is exact for the corresponding discrete, held-sample inertial model with correct biases and initial tilt.

For the algebra and a restricted bias extension, use

$$p(t) = c(t) + R_z(\psi)d(t), \qquad c(t) = \sum_{j=0}^{K} c_j \tau^j. \tag{4}$$

The primary initializer uses $K = 1$. Under constant attitude, a constant body accelerometer-bias error produces a constant world acceleration error and is absorbed by $K = 2$. This extension is exact only under that condition. With varying attitude, the correct sensitivity is a double integral of $Q(t)$, rather than a scalar quadratic term; the corresponding model is discussed in Section VIII.

IV. A Profiled Yaw Candidate

A. Exact Lift of the Squared Range

Temporarily consider noise-free ranges. Define

$$y_i = \tfrac{1}{2}\left(r_i^2 - \|a_i\|^2 - \|d_i\|^2\right) + a_{iz}d_{iz}. \tag{5}$$

Expanding (1)–(4) yields

$$y = B\nu + Dc + Cu, \qquad u = \begin{bmatrix}\cos\psi & \sin\psi\end{bmatrix}^\top. \tag{6}$$

Here c stacks the $K+1$ three-vectors c_j. The row blocks are

$$B_i = \left[\{\tau_i^\ell\}_{\ell=0}^{2K}, \{\tau_i^j d_{ix}, \tau_i^j d_{iy}\}_{j=0}^{K}\right], \tag{7}$$

$$D_{ij} = \tau_i^j \begin{bmatrix}-a_{ix} & -a_{iy} & d_{iz} - a_{iz}\end{bmatrix}, \tag{8}$$

$$C_i = \begin{bmatrix}-a_{ix}d_{ix} - a_{iy}d_{iy} & a_{ix}d_{iy} - a_{iy}d_{ix}\end{bmatrix}. \tag{9}$$

The first $2K+1$ components of ν are the coefficients of $\frac{1}{2}\|c(t)\|^2$. Its remaining components are the horizontal entries of $R_z(\psi)^\top c_j$. The vertical cross terms are retained in D because a yaw rotation leaves c_{jz} unchanged. Thus ν has $4K+3$ components, and $[\nu^\top, c^\top]^\top$ has $7K+6$. For $K = 1$ and $K = 2$, the complete lifts contain 15 and 22 scalar coordinates, respectively, including the two entries of u.

Equation (6) is an exact algebraic identity on the physical parameter set. Treating ν as independent of c and u enlarges that set. This distinction matters: a minimizer of the relaxed squared-range problem need not minimize the physically constrained raw-range problem.

B. Eliminating Linear Nuisances

Let W be a positive diagonal whitening matrix and define

$$\begin{aligned} L &= W[B\ D], \quad P_L = I - LL^\dagger, \\ \bar{y} &= P_L W y, \qquad \bar{C} = P_L W C. \end{aligned} \tag{10}$$

The analysis uses the exact column space of L. The implementation retains SVD directions with $s_j > 10^{-10} s_{\max}$ and applies the complement without storing a dense $N \times N$ matrix. Eliminating the linear variables gives

$$\min_{\|u\|=1} \left\|\bar{y} - \bar{C}u\right\|^2 = \min_{\|u\|=1} u^\top H u - 2h^\top u + \text{const}, \quad (11)$$

where $H = \bar{C}^\top \bar{C}$ and $h = \bar{C}^\top \bar{y}$. The model dimension is fixed for a chosen K; increasing the number of transactions adds rows rather than optimization states.

Set $\Delta = H_{22} - H_{11}$ and $q = \tan(\psi/2)$. The stationary-angle equation becomes

$$\begin{aligned} 0 = {} & (H_{12} + h_2)q^4 + 2(h_1 - \Delta)q^3 - 6H_{12}q^2 \\ & + 2(h_1 + \Delta)q + (H_{12} - h_2). \end{aligned} \quad (12)$$

We enumerate the real roots, map them back to the circle, include $\psi = \pi$ to cover the point omitted by the half-angle chart, and evaluate (11). Degenerate constant objectives are handled directly. This enumeration finds the global minimum of the two-dimensional relaxed candidate problem; raw-range refinement is a subsequent local solve.

Proposition 1 (Sufficient exact yaw recovery): Assume noiseless ranges generated by (4), an exact inertial displacement, and a positive diagonal W. If $\operatorname{rank}(\bar{C}) = 2$, the global minimizer of (11) is the true u, and hence the true yaw modulo 2π. Given that yaw, translation is unique if

$$\operatorname{rank}(P_{WB} W D) = 3(K+1), \qquad P_{WB} = I - (WB)(WB)^\dagger. \quad (13)$$

Proof: The true physical variables satisfy (6), so their projected residual is zero. Any global minimizer therefore satisfies $\bar{C}(u - u_\star) = 0$. Full column rank implies $u = u_\star$. With u fixed, eliminating ν leaves a linear system in c whose coefficient matrix is $P_{WB} W D$; (13) implies a unique solution. ∎

These are sufficient conditions: the relaxation may discard information contained in the nonlinear consistency constraints.

C. Noise Treatment and Raw-Range Refinement

For Gaussian inlier noise in (1), replace r_i^2 in (5) by $r_i^2 - \sigma_i^2$. The remaining random part is

$$\epsilon_i = \rho_i e_i + \tfrac{1}{2}(e_i^2 - \sigma_i^2), \quad \mathbb{E}[\epsilon_i] = 0, \quad \operatorname{var}(\epsilon_i) = \rho_i^2 \sigma_i^2 + \tfrac{1}{2}\sigma_i^4. \quad (14)$$

The true ρ_i is unavailable. The implementation uses

$$W_{ii} = \left[\max(r_i, \sigma_i)^2 \sigma_i^2 + \tfrac{1}{2}\sigma_i^4\right]^{-1/2}. \quad (15)$$

This plug-in whitening approximates the squared-range variance. Robust refinement then operates directly on the original range residuals.

Accordingly, the lift supplies a yaw candidate, after which we minimize

$$J(c, \psi) = \sum_i \rho_H\left(\frac{\|c(t_i) + R_z(\psi) d_i - a_i\| - r_i}{\sigma_i}\right), \quad (16)$$

where ρ_H is the Huber loss with normalized threshold 2. For $K = 1$, robust constant-velocity localization supplies the translation seed, because recovering translation from the relaxed lift can be ill conditioned. The analytic Jacobian includes

$$J_{c_j,i} = \frac{\tau_i^j n_i^\top}{\sigma_i}, \qquad J_{\psi,i} = \frac{n_i^\top [e_3]_\times R_z(\psi) d_i}{\sigma_i}, \quad (17)$$

where $n_i = (p(t_i) - a_i)/\|p(t_i) - a_i\|$ is the unit range direction. All fits use trust-region reflective least squares, analytic Jacobians, Huber threshold 2, a cap of 120 function evaluations, and objective, step, and gradient stopping tolerances of 10^{-8}. These limits apply to each solve, not to the total work of a multi-start method.

TABLE I
PROFILED YAW CANDIDATE AND REFINEMENT ($K = 1$).

Input: calibrated ranges, anchors, scales, gravity-aligned IMU.
1. Integrate and reanchor $d, \dot{d}$ at the endpoint.
2. Refine CV translation from linear squared-range and centroid/zero-velocity seeds; retain the lower raw cost.
3. Form W, y, L, C; apply the SVD complement and form H, h. Normalize quartic coefficients; zero relative magnitudes $\le 10^{-13}$.
4. Keep roots with $|\operatorname{Im} q| < 10^{-7}(1 + |\operatorname{Re} q|)$; map to yaw, add $0, \pi$, deduplicate within 10^{-7} rad, and rank circle costs.
5. Refine the best yaw with the CV translation on original ranges. Additional yaws repeat only this step.

Degeneracy: constant quartics retain auxiliary angles; rank-deficient cases proceed to raw refinement.

V. Bias and Heading Identifiability

Partition a weighted raw-range Jacobian into nuisance and yaw columns, $A = [A_n \; a_\psi]$. A local information measure after nuisance elimination is

$$\mathcal{I}_{\psi|n} = \left\|(I - A_n A_n^\dagger) a_\psi\right\|^2. \quad (18)$$

With final Huber iteratively reweighted least-squares (IRLS) weights, $A = \operatorname{diag}(\sqrt{w_i}) J$ gives a local surrogate information matrix. Equation (18) measures the yaw component outside the nuisance span (Fig. 1c), in inverse squared radians. Including bias-sensitive columns evaluates heading information with that bias freedom.

Proposition 2 (Constant-acceleration ambiguity): Suppose attitude is constant, the gyroscope bias is known, and the measured specific force f_m is constant over a window. If the constant accelerometer bias is unrestricted, position-only ranges cannot identify yaw, even with nondegenerate surveyed anchors and noiseless, arbitrarily frequent measurements.

Proof: Let a solution have $R = R_z(\psi) Q$ and bias b_a. For any candidate yaw ψ', define

$$b_a' = f_m - Q^\top R_z(\psi')^\top R_z(\psi) Q (f_m - b_a). \quad (19)$$

Because Q and f_m are constant, b_a' is constant. Substitution gives $R_z(\psi') Q (f_m - b_a') = R_z(\psi) Q (f_m - b_a)$. The acceleration, and therefore the entire position trajectory with the same initial position and velocity, is unchanged. All zero-lever-arm ranges are identical for this continuum of yaw–bias pairs. ∎

For the quadratic translation model, the same effect appears in (18): the yaw sensitivity of a quadratic inertial displacement lies in the span of the quadratic translation columns. Its projected information is zero. With varying acceleration, the nonquadratic component can retain heading information after constant-bias elimination. Useful excitation separates this informative component from measurement and model error.

Proposition 3 (Polynomial nuisance invariance): Let all translation coefficients in (4) be unrestricted. Replacing $d(t)$ by $d'(t) = d(t) + b(t)$, where $b(t) = \sum_{j=0}^{K} b_j \tau^j$, leaves the raw-range objective profiled over translation unchanged for every yaw. It also leaves the nuisance-projected local yaw information unchanged at corresponding states. The relaxed circle objective is unchanged when W depends only on the unchanged ranges and their scales.

Proof: For every ψ, set $c'_j = c_j - R_z(\psi) b_j$. Then $c'(t) + R_z(\psi) d'(t) = c(t) + R_z(\psi) d(t)$, so corresponding residuals and any residual-based robust weights are identical. The change in the yaw Jacobian is a linear combination of the translation columns and is removed in (18). For the lift, the column space of B is unchanged because $\tau^j b_x$ and $\tau^j b_y$ have degree at most $2K$. Changes in D lie in that space; changes in C lie in the space of $[B\ D]$; and the change in y is a combination of the same polynomial and cross-term columns. Projection therefore leaves $\bar{y}$ and $\bar{C}$ unchanged. ∎

The invariance identifies an equivalence class of displacement errors absorbed by translation. Constant world acceleration bias is one such quadratic contamination.

Proposition 4 (Bounded-rotation approximation): Let Q be absolutely continuous on $[t, T]$ with $\left\|\dot{Q}(s)\right\|_2 \le \Omega$ almost everywhere, and define the endpoint-reanchored bias sensitivity

$$B_e(t) = \int_t^T (s - t) Q(s)\, ds. \tag{20}$$

Then

$$\left\| B_e(t) - \tfrac{1}{2}(T-t)^2 Q(T) \right\|_2 \le \frac{\Omega}{6}(T-t)^3. \tag{21}$$

Consequently, a constant body-bias perturbation δb_a is absorbed by the $K = 2$ model up to a displacement remainder, and hence a raw-range perturbation, bounded by $\Omega \left\|\delta b_a\right\| |t - T|^3/6$.

Proof: Absolute continuity gives $\left\|Q(s) - Q(T)\right\|_2 \le \Omega(T - s)$. Integrating $(s - t)\Omega(T - s)$ from t to T gives (21). Multiplication by δb_a and a yaw rotation preserves the bound, and the reverse triangle inequality bounds the resulting range change. ∎

This approximation quantifies how rotation rate, bias magnitude, and window duration control the residual displacement mismatch.

An isolated initial tilt error has a related structure. With exact subsequent gyroscope increments and specific force, $\hat{Q}(t) = EQ(t)$ for constant E, giving $\hat{d}(t) = Ed(t) + \frac{1}{2}\tau^2(I - E)g$. Quadratic translation absorbs the gravity term exactly; the remaining shape error is bounded by $2\sin(\theta_E/2)\left\|d(t)\right\|$, where θ_E is the rotation angle of E. Gyroscope errors introduce the additional time-varying component of E.

VI. Experimental Protocol

A. Paired Candidate and Computation Comparison

The candidate comparison uses two far-field geometries: an endpoint 30 m horizontally from the anchor centroid, eight anchors at $(\pm 4, \pm 4, z)$ m with $z \in \{0, h\}$, and $h = 3$ or 0.1 m. The endpoint height is 1.2 m; endpoint azimuth and yaw are independently uniform, and each velocity component is uniform on $[-0.5, 0.5]$ m/s. Each four-second window has 240 sorted random transaction times, independently selected anchors, and 0.10 m Gaussian range noise. The endpoint-reanchored, calibrated inertial shape is

$$d(\tau) = \begin{bmatrix} 0.8\sin(1.7\tau) - 1.36\tau \\ 0.6(\cos(1.1\tau) - 1) \\ 0.15\sin(0.7\tau) - 0.105\tau \end{bmatrix}, \quad \tau \in [-4, 0]. \tag{22}$$

It satisfies $d(0) = \dot{d}(0) = 0$. We use 200 predetermined seeds per geometry for paired comparisons.

Local uses yaw zero; Profile-1 uses the best circle candidate; Uniform-2 and Uniform-4 use two and four evenly spaced yaw seeds; Uniform-12 is a finite numerical reference. Every method recomputes the identical two-solve CV initializer and uses its selected translation, the same Huber objective, and the optimizer settings in Table I. One yaw candidate means one seven-variable refinement. Equal seed counts are not equal computation: we report total function evaluations across both CV solves and every refinement, plus sequential wall time including CV, projection, root enumeration, and final curvature construction. The fixed cap is 120 evaluations *per solve*, so uniform multi-start methods have a larger total allowance.

An objective miss means

$$J - J_{\text{best}} > \max\{10^{-3}, 10^{-5}|J_{\text{best}}|\}, \tag{23}$$

where J_{best} is the best achieved objective among the five methods on that identical window. A finite multi-start reference can itself miss this value. Position and wrapped heading errors use synthetic truth; they are separate outcomes from this optimization comparison.

Sequential timing uses the first eight predetermined seeds per geometry and five interleaved repeats per method, after an excluded warm-up. This gives 400 timed calls on one Windows Intel Core i5-12600KF desktop, one BLAS thread, with no concurrent experiment fits. The environment is Python 3.11.15, NumPy 1.26.4, and SciPy 1.17.1. Timings describe this desktop subset, not embedded latency or a fixed-time optimality guarantee.

B. Model-Order and Calibration Controls

The model-order controls include near-field calibrated motion, constant-world acceleration contamination, and physical 200 Hz IMU propagation with combined accelerometer-bias, gyroscope-bias, and initial-tilt errors. The latter uses 30 m stand-off, 3 m anchor height span, and varying roll and pitch. Ideal, Mild, and Moderate use component standard deviations $(\sigma_a, \sigma_g, \sigma_{\text{tilt}})$ of $(0, 0, 0)$, $(0.02, 0.002, 0.5^\circ)$, and $(0.10, 0.01, 2^\circ)$, respectively, with acceleration in m/s^2 and

gyro rate in rad/s. Each seed draws independent zero-mean Gaussian components for constant three-axis biases and an initial body-right roll/pitch rotation vector; the same draws are scaled across levels. No IMU white noise is added. A matched $K = 1/K = 2$ comparison uses 200 paired seeds per level. Both models refine two *translation* initializations at their respective single lifted yaw: recovered lifted polynomial coefficients, and the common CV position/velocity with zero quadratic coefficient when needed. Thus 1,200 fits control the outer seed policy while changing polynomial order and its induced lift. Objectives are not ranked across motion models of different order.

C. Real Windows and Statistical Units

We use TU Delft dataset version 1 [11], with two recordings each of triangle, octagon, hourglass, random, star, and square motion. The earlier recording supplies per-anchor bias medians and MAD scales, clipped to $[0.08, 0.50]$ m; the later recording is evaluated. All recordings are from one date and have prior development exposure. Truth supplies offline training calibration, valid-interval selection, a common 0.20 s IMU/UWB-to-OptiTrack alignment, and scoring. The online initializer consumes ranges and IMU samples; the recordings form a development-data evaluation.

Each evaluation log supplies eleven windows: 2 and 4 s ending at 5 s, and 2, 4, and 8 s ending at 10, 20, and 30 s. Times use the retained clock after the approximately 1.5 s static prefix; a requested endpoint uses the last arrived sample at or before that time. We use all eight anchors or the lowest four by surveyed height. Four equivalent inertial yaw frames replace d by $R_z(\phi)d$, $\phi \in \{0, \pi/2, \pi, 3\pi/2\}$, while the yaw coordinate changes by $-\phi$. The world translation, anchors, and truth are unchanged. Hence 528 configurations represent 66 overlapping physical time windows, 132 layout-specific windows, and six logs; coordinate transformations are not independent acquisitions.

We compare six policies: Local, Profile-1, the first two distinct circle candidates (Profile-2), all distinct circle/auxiliary candidates (Profile-all, at most six), Uniform-4, and Uniform-12. Each uses the same CV seed and affine model. Here a miss exceeds the best observed six-method objective by $10^{-4}(1 + J_{\text{best}})$, following the existing real-window protocol. We summarize results within each recording. As $d(T) = 0$, the fitted coefficient c_0 is directly the endpoint position in the anchor/OptiTrack frame. Its Euclidean error uses truth interpolated at the actual endpoint in this world frame.

Saved endpoints and function counts provide physical errors and computation for all 3,168 fits. The supplement reports the historical real-window timings separately from the candidate-cost desktop experiment.

VII. Results

A. Candidate Quality versus Actual Computation

With a 3 m anchor height span, Profile-1 reaches the best observed objective in all 200 trials and reduces position/heading 95th-percentile errors from Local's 4.462 m/$155.54°$ to 0.509 m/$3.48°$ (Table II). Uniform-2 matches this accuracy. Profiling requires 67 median function evaluations versus 91 for Uniform-2, with median times of 47.1 versus 66.6 ms: a 29.4% reduction. Figure 2 visualizes the full cost–miss tradeoff.

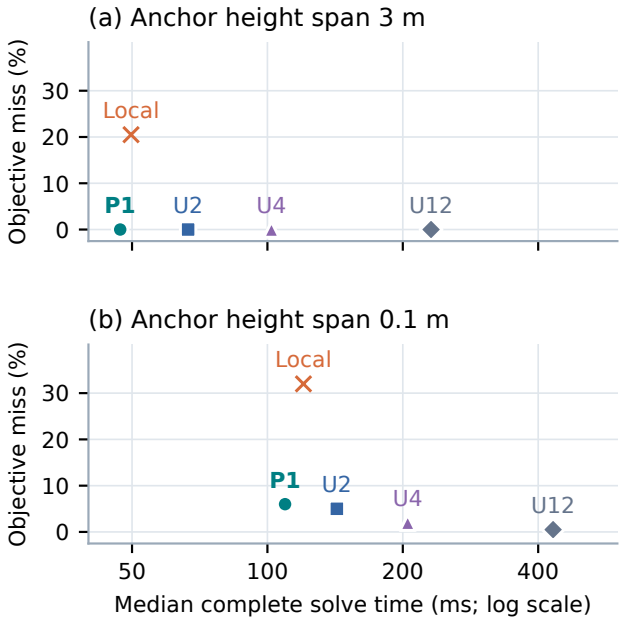


Fig. 2. Optimization reliability–computation tradeoff for the two far-field geometries. P1 denotes Profile-1 and Uk denotes Uniform-k. Costs include common CV translation, candidate generation, and raw-range refinement. Each miss rate uses 200 paired windows; times use the separate sequential subset on a logarithmic axis.

Weak vertical geometry changes the tradeoff. At 0.1 m height span, Profile-1 retains 12 objective misses; Uniform-2 and Uniform-4 reduce these to ten and four, and Uniform-12 still misses once. Six Local fits reach the per-solve evaluation cap. Additional starts improve objective reliability at greater cost, while position-error tails near 4 m reveal the remaining height ambiguity.

B. Nuisance Absorption Trades Model Error for Information

In the matched model-order comparison (Table III), $K = 2$ increases error tails under ideal input but reduces them under the tested combined calibration perturbations. At the moderate level, position and physical endpoint-heading 95th percentiles decrease from 4.535 m and $54.85°$ to 0.761 m and $12.39°$. This improvement is consistent with the bounded-rotation approximation.

For the constant-acceleration control, a heading passes the information diagnostic when $\mathcal{I}_{\psi|n} \geq 1/(0.35\,\text{rad})^2$; no residual gate is applied to this count. With free bias, all 1,000 cases are rejected despite median position error of 0.067 m. With bias fixed and a component perturbation scale of $0.3\,\text{m/s}^2$, 992 cases pass but 554 have heading error above 0.35 rad. This diagnostic illustrates the role of bias freedom; actual adaptive-window handoff also checks residuals, as specified in the supplement.

TABLE II

MATCHED CANDIDATE POLICIES AT 30 M STAND-OFF: 200 PAIRED SEEDS PER GEOMETRY. MISSES USE THE FIVE-METHOD BEST OBSERVED OBJECTIVE. ERRORS ARE 95TH PERCENTILES. EVALUATIONS INCLUDE BOTH CV SOLVES AND ALL YAW REFINEMENTS. WALL TIME IS MEDIAN/95TH PERCENTILE FROM THE SEPARATE 8-SEED-PER-GEOMETRY, 5-REPEAT SEQUENTIAL SUBSET.

Height (m)	Policy	Miss / 200	$e_{p,95}$ (m)	$e_{\psi,95}$ (°)	Median evals	Time med. / 95th (ms)
3	Local	41	4.462	155.54	70.5	49.7 / 86.1
	Profile-1	0	0.509	3.48	67	47.1 / 60.1
	Uniform-2	0	0.509	3.48	91	66.6 / 89.7
	Uniform-4	0	0.509	3.48	132	102.1 / 166.7
	Uniform-12	0	0.509	3.48	290	231.1 / 334.4
0.1	Local	64	6.870	152.54	126	120.3 / 252.9
	Profile-1	12	3.946	4.21	121.5	109.5 / 218.6
	Uniform-2	10	3.946	4.21	165.5	142.7 / 325.0
	Uniform-4	4	3.946	4.21	233	205.1 / 385.4
	Uniform-12	1	3.981	4.21	498	431.7 / 619.8

TABLE III

MATCHED POLYNOMIAL ORDERS: 200 PAIRED SEEDS PER LEVEL, TWO OUTER TRANSLATION INITIALIZATIONS PER MODEL AND ONE LIFTED YAW PER MODEL. ENTRIES ARE POSITION/PHYSICAL-HEADING 95TH PERCENTILES (M/°).

Calibration	$K = 1$	$K = 2$
Ideal	0.546 / 3.58	0.732 / 4.75
Mild	1.031 / 16.90	0.731 / 5.52
Moderate	4.535 / 54.85	0.761 / 12.39

C. Real Objectives and Endpoint Position Are Different Outcomes

Across the 528 configurations, Profile-1 more than halves objective misses, from 87 to 40, while reducing total function evaluations from 36,157 to 32,632. It alone reaches the reference in 58 paired configurations, versus eleven for Local alone. Additional candidates reduce misses to 28 for Profile-2 and eleven for Profile-all; Uniform-4 and Uniform-12 give twelve and zero.

Figure 3 pairs optimization and physical outcomes by log. Pooled endpoint-error median/95th percentile changes from Local's $0.680/4.241$ m to Profile-1's $0.658/4.084$ m; endpoint RMSE decreases from 1.996 to 1.898 m. Uniform-12 gives $0.658/4.241$ m and 1.926 m, respectively. Position gains vary by recording: hourglass and random have higher Profile-1 position tails despite fewer objective misses.

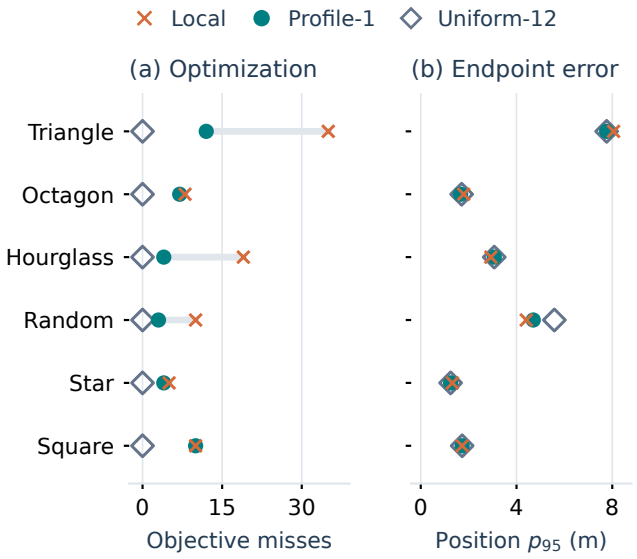


Fig. 3. Real-window outcomes by recording: objective misses and endpoint position 95th percentiles. Each recording supplies 88 correlated configurations from eleven physical windows, two anchor layouts, and four equivalent yaw frames. Connecting segments compare Local and Profile-1; diamonds show Uniform-12.

D. Integration into the Common Invariant Tracker

The fitted endpoint initializes a common 15-state InEKF, using physical-to-group covariance conversion and each capture range once. After static IMU alignment, the initializer acquires a 2 s window. The supplement defines the handoff, gates, and time origins.

In six continuous-ranging replays, CV, Local, Uniform-12, and Profile-1 each reach five stable successes and median trajectory RMSE of approximately 0.285 m. With a ten-second outage, Profile-1 gives maximum available-output error 27.72 m and median RMSE 2.16 m, versus 17.65 m and 1.39 m for the earlier lifted capture. Local and Uniform-12 have essentially the same outage outcomes as Profile-1. A better initialization objective therefore does not ensure better subsequent tracking. Handoff covariance, heading confidence, and acquisition supervision may affect these transients; the earlier capture also differs in its warm-up and output policy, so these comparisons do not isolate a cause.

VIII. SCOPE AND OUTLOOK

The recovery analysis applies to the exact window model and projector; the implementation uses truncated SVD and local raw-range refinement. For a rotating body, constant accelerometer-bias sensitivity takes the form

$$d_{\mathrm{true}}(t) = d_{\mathrm{nom}}(t) - B_a(t)\delta b_a, \tag{24}$$

where B_a is the endpoint-reanchored double integral of Q. Equation (21) guides the quadratic approximation through

rotation rate and window duration. The calibration controls exercise combined bias and tilt errors; IMU white noise and timing uncertainty are further extensions.

The real study uses six same-date development recordings, correlated windows/layouts/frames, and truth-assisted offline calibration and alignment. It evaluates position and optimization performance. Independent recordings with heading truth, covariance calibration, and a matched external initializer are the next steps toward wider deployment.

IX. Conclusion

Polynomial nuisance profiling turns the unknown yaw of an inertial trajectory shape into a compact unit-circle candidate problem. At 30 m stand-off with a 3 m anchor height span, a single profiled candidate matches the observed objective quality of two uniform starts, with 29.4% lower median solve time on the sequential desktop timing subset. The invariance, ambiguity, and rotation-bound analysis explains how polynomial order trades model flexibility for heading information. Real-window comparisons further show a substantial reduction in objective misses and modest, recording-dependent position gains. Together, the method and analysis provide a practical initialization component for UWB–IMU invariant tracking.

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